SMARTBRIDGE
Let's Bridge the Gap

SKILL WALLET
A SmartBridge Product

Coding & Solution

Date	29 June 2026
Team ID	N/A
Project Name	Rising Waters
Maximum Marks	5 Marks

Coding & Solution

This section evaluates the quality and completeness of your implemented code and the overall solution. Your submission should demonstrate working functionality, clean code practices, and alignment with your defined requirements.

Instructions:

1. Ensure your code is well-structured, commented, and follows best practices for your chosen technology stack.
2. The solution should address the core problem statement and implement the key functional requirements.
3. Include all source files, configuration files, and setup instructions needed to run the application.
4. Attach or link to your code repository (e.g., GitHub) in the table below.

Solution Summary

Field	Details
Repository Link / URL	https://github.com/venkatvenkat34567-eng/Rising-Waters-Project-.git
Programming Language(s)	Python
Framework(s) Used	Flask, Scikit-learn, Pandas, NumPy, Matplotlib
Key Features Implemented	• Flood prediction using Machine Learning • Water level monitoring • Rainfall data analysis • Flood risk classification (Low, Medium, High) • Data visualization using graphs • User-friendly prediction interface
Pending / Incomplete Features	• Live IoT sensor integration • Real-time weather API integration • SMS/Email alert system • Mobile application support

Setup / Run Instructions	1. Install Python 3.10+ 2. Install dependencies using `pip install -r requirements.txt` 3. Run `python app.py` 4. Open `http://localhost:5000` in a browser 5. Enter rainfall and water-level values to predict flood risk

Code Quality Checklist

S.No	Criteria	Status (Yes / No)
1	Code is modular and organized into functions / classes	Yes
2	Meaningful variable and function names are used	Yes
3	Code includes comments / documentation where necessary	Yes
4	Error handling is implemented for critical operations	Yes
5	The application runs without critical errors	Yes
6	Code is committed to a version control repository	Yes

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